

2022



Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer all sections as directed.

Section-A

Objective type Questions

1. Choose the correct alternative for each of the
following questions : 2×10=20

(a) Process of inserting an element in stack
is called _____

(i) Create

~~(ii)~~ Push

(ii) Evaluation

(iv) POP

(b) Linked list is considered as an example of ____ type of memory allocation.

~~(i)~~ Dynamic

(ii) Static

(iii) Compile time

(iv) Heap

(c) The essential condition which is checked before insertion in a linked queue is :

(i) Underflow

~~(ii)~~ Overflow

~~(iii)~~ Front value

(iv) Rear value

(d) Process of removing an element from queue is called _____

- (i) Create
- (ii) Push
- (iii) Insert
- ☒ (iv) Pop

(e) What is the value of the postfix expression $6324 + - *$:

- (i) 1
- (ii) 40
- (iii) 74
- ☒ (iv) -18

(f) In linked list implementation a node carries information regarding :

- (i) Data

(ii) Link

~~(iii) Data and link~~

(iv) Node

(g) Assuming int is of 4 byte, what is the size of intarr[15] :

(i) 15

(ii) 19

(iii) 11

~~(iv) 60~~

(h) Which of the following sorting algorithm can be used to sort a random linked list with minimum time complexity :

(i) Insertion sort

(ii) Quick sort

~~(iii)~~ Heap sort

~~(iv)~~ Marge sort

(i) In general the index of the first element in an array is ____

☒ (i) 0

(ii) -1

(iii) 2

(iv) 1

(j) The post fix forms of $A * B + C / D$:

(i) $*AB / CD +$

☒ (ii) $AB * CD / +$

(iii) $A * BC + / D$

(iv) $ABCD + / *$

Section-B

Short Answer type Questions

Answer any **four** questions of the following :

5×4=20

2. Explain the various types of data structure.
3. Explain selection sort algorithm.
- * 4. Mention the various application of stack.
5. Write a program to insert 10 elements in an array.
6. Explain BST.
7. Explain BFS and DFS.
8. What is queue ? Mention its application.

Section-C

Long Answer type Questions

Answer any **two** questions of the following :

15×2=30

9. What is linked list ? Briefly define its types with example.

10. Write short notes on the following :

- (a) Inorder traversal
- (b) Preorder traversal
- (c) ~~Post~~ Inorder traversal

11. What is a pointer ? How dynamic memory is allocated ?

12. How to convert infix notation into postfix notation ?

13. Write an algorithm to insert a node at the beginning of a linked list.

_____ x _____